



Do Airline Crashes Move Markets?

We link NTSB accident data with airline & manufacturer stock prices to measure how much markets react to crashes.

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Roadmap

01	02	03
Situation	Complication	Resolution
Airline crash data and market reaction data collection	Does the market punish airlines & manufacturers?	Event-study methodology and key insights

Situation

Data Sources

- NTSB accident data (extraction, cleaning, severity classification)
- Market reaction data (stock prices, SPY benchmark, ticker mapping)

Understanding

- Stock reaction trend is unclear and often masked by market movement
- Magnitude of reaction & recovery timeline is unclear



Complication

Market Punishment

Does the market punish airlines & manufacturers after crashes?

Duration & Magnitude

How much and for how long is the market reaction?

Investor Response

How should investors react to airline crashes?

Company Management

How can involved companies manage market reaction?

Resolution



Event-Study Methodology

Returns → abnormal returns → event window



Key Insights

Impact, worst drop, recovery patterns



Future Directions

Limitations and areas for expansion

Project Files



ntbs.ipynb

Crash data extraction, cleaning &
severity labeling



stocks.ipynb

Stock price loading, ticker mapping &
event alignment



final.ipynb

CAR, MAE, TTR computation & all final
results

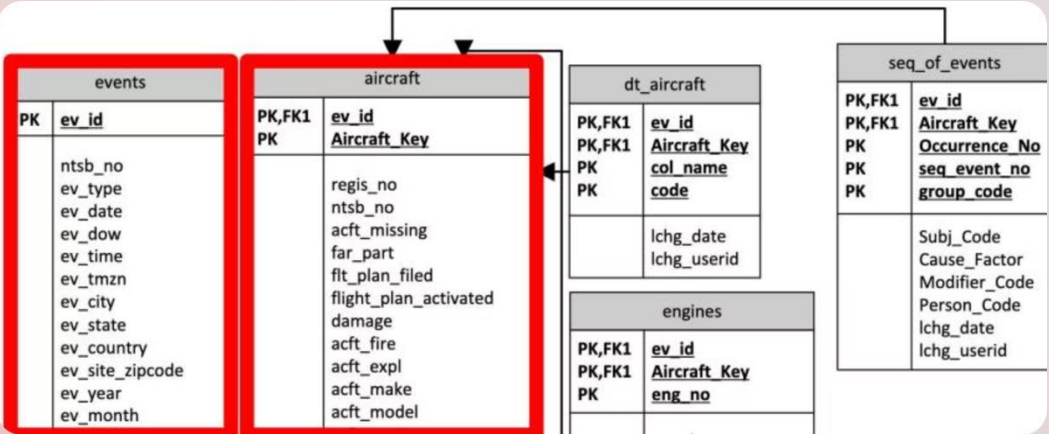


dashboard.py

Streamlit dashboard

What We're Analyzing

NTSB Serious Airline Accidents (2008–2025)



263

Total Crashes

Filtered for >1 total
injuries

2

Manufacturers

Boeing & Airbus in scope

17

Years Analyzed

2008/02/08 to
2025/09/30

Source: <https://www.ntsbn.gov/>

Technical steps: Decompressed large database into 10+ tables, picked relevant fields, merged tables & cleaned data

Crash Data Module (NTSB)

What's in This File

- Raw NTSB crash tables (events, aircraft, injuries)
- Cleaned crash records (dates, operators, manufacturers)
- Total injuries + severity categories
- Output dataset: processed.parquet

How It Works

- Extracts multiple NTSB CSV tables
- Merges on event ID (ev_id)
- Cleans dates, removes invalid entries
- Computes severity labels from injury totals



Why This Matters

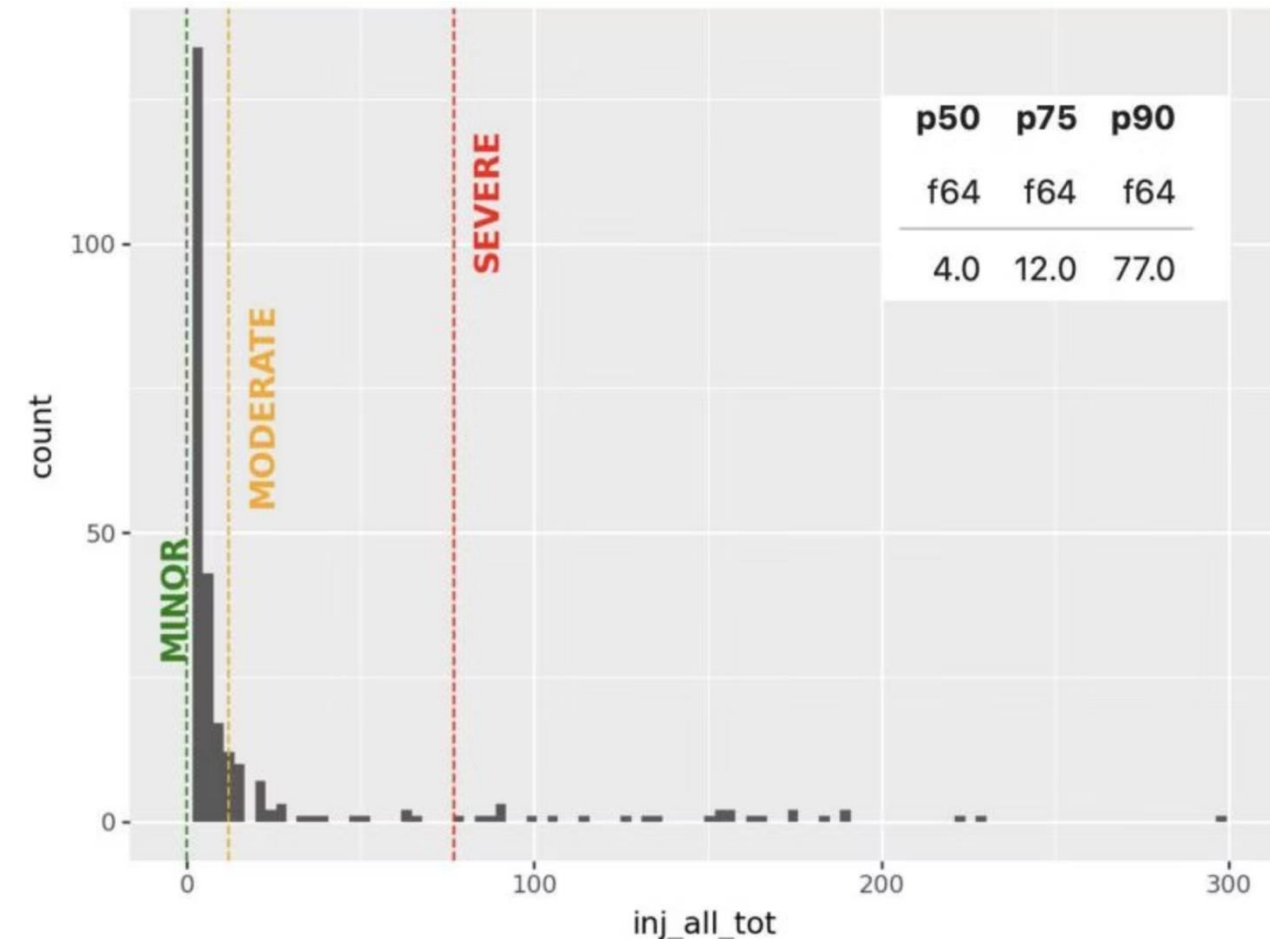
Creates the crash event list used in the event study, defines severity groups for MAE/TTR comparisons, and provides the cleaned dataset for stocks.py and final.ipynb.

Most Crashes Are Minor, A Few Are Catastrophic

Injury Buckets

- **Minor:** ≤ 12 injuries (203 events)
- **Moderate:** 12–77 injuries (34 events)
- **Severe:** > 77 injuries (26 events)

The distribution is highly skewed, with the vast majority of events having minimal injuries. Only a small fraction represent catastrophic accidents.



Accidents Over Time by Manufacturer

What

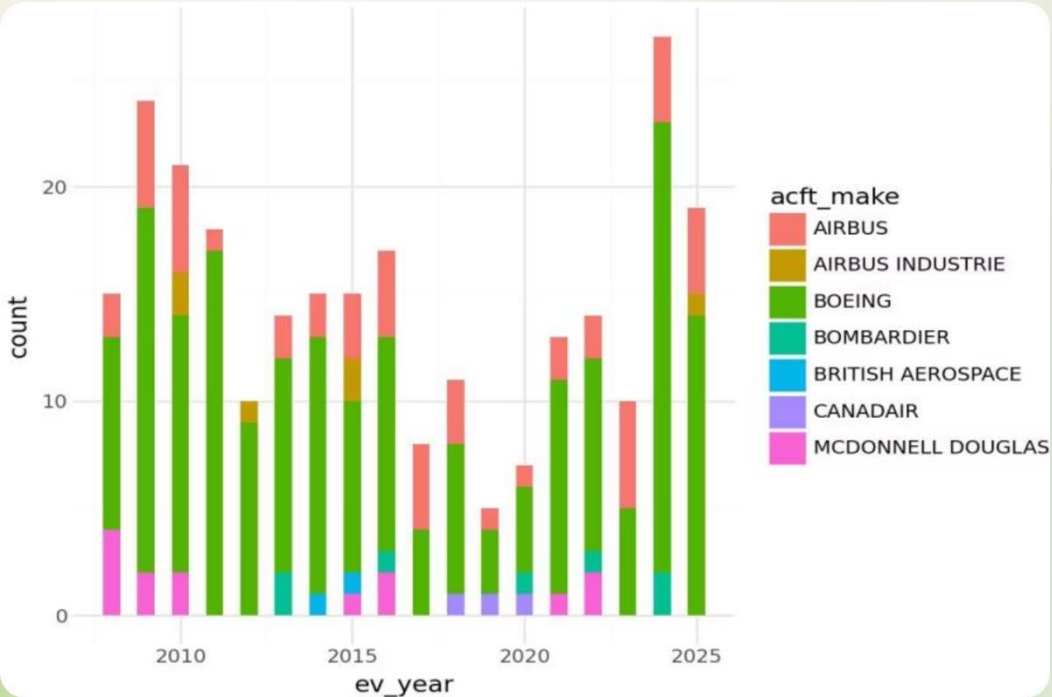
Yearly crash counts by manufacturer

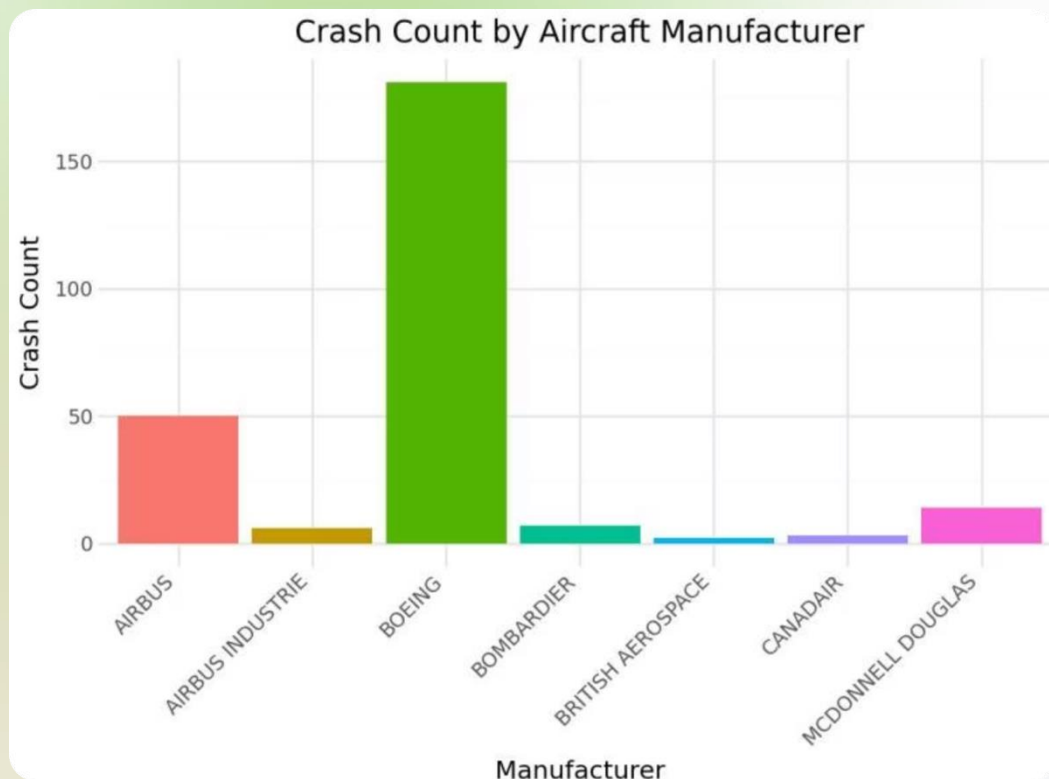
Why

See trends & manufacturer dominance

How

Grouped by year + manufacturer



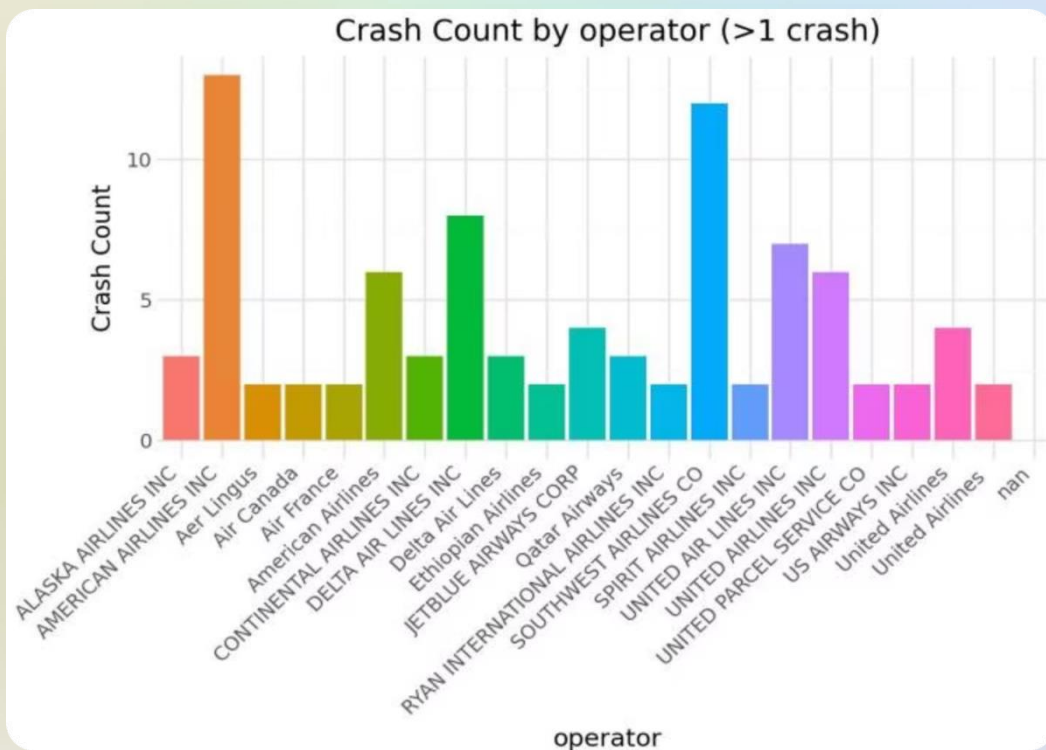


Crash Count by Manufacturer

What: Crash counts for each aircraft manufacturer show which manufacturers appear most in the crash dataset.

Why: Boeing and Airbus dominate the dataset, reflecting their market share in commercial aviation.

How: Counted occurrences of each manufacturer in the cleaned NTSB data.



Crash Count by Operator

Airlines with More Than One Recorded Crash

This chart highlights repeat-event operators and shows how crashes are distributed across airlines. United Airlines and Delta Air Lines lead in crash frequency, though this partially reflects their large operational scale.

Catastrophic Crashes in the Data

A few events account for a large share of total injuries in the sample. These are headline events with the strongest stock-market reaction potential.

Operator	Manufacturer	Injuries	Year	Date
Malaysian Airlines	Boeing	298	2014	2014-07-17
Air France	Airbus	228	2009	2009-06-01
Unknown	Airbus	224	2015	2015-10-31
Unknown	Boeing	191	2018	2018-10-29
Asiana Airlines	Boeing	190	2013	2013-07-06

From Crash Data to Public Stocks

Mapping Criteria

We only use crashes where we can map:

- Aircraft Manufacturer → stock ticker
- Operator → stock ticker

Price Data Sources

- Sourced from Yahoo Finance (yfinance library)
- Daily adjusted closing price per ticker
- Daily adjusted closing price per market index

93

Total Events

Crashes with ticker data

73

Minor

12

Moderate

8

Severe

Stock Data Module

stocks.py

Load Prices

Retrieves daily prices for airlines & manufacturers from Yahoo Finance

1

2

Compute Returns

Merges prices with crash events and computes stock vs. market returns

3

Align Events

Aligns crash dates to trading days and maps NTSB operators to stock tickers

📌 **Why This Matters:** Provides stock + market returns for CAR, MAE, TTR calculations. Connects crash events to financial reactions and supplies the aligned dataset for the event-study notebook.

What Are We Trying to Measure?



Impact Size

How large are the abnormal returns following a crash?



Trend Over Time

How does the stock react in the short-term following the event?



Recovery Time

How long does it take the stock to recover to pre-crash levels?



Heterogeneity

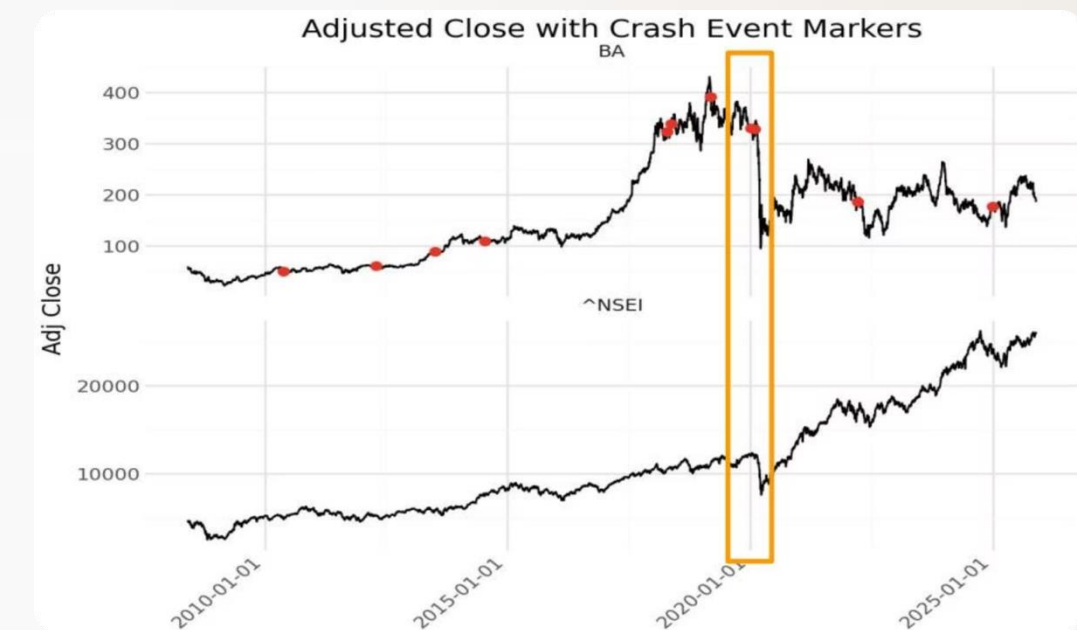
Does impact differ by region, year, manufacturer, or severity?

Why Raw Stock Prices Mislead

What: Long-run stock prices for Boeing (BA) and Airbus (EADSY) with crash dates marked as red dots.

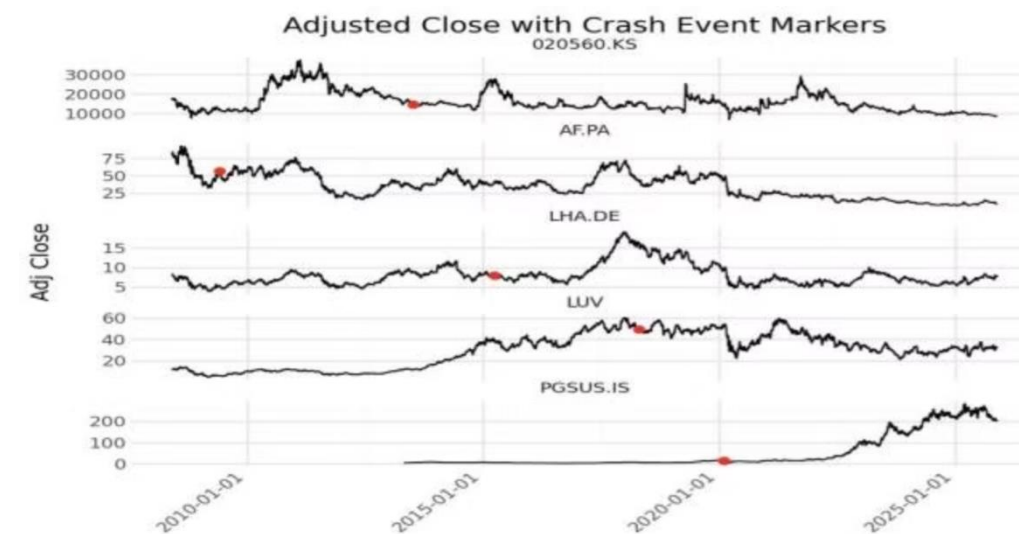
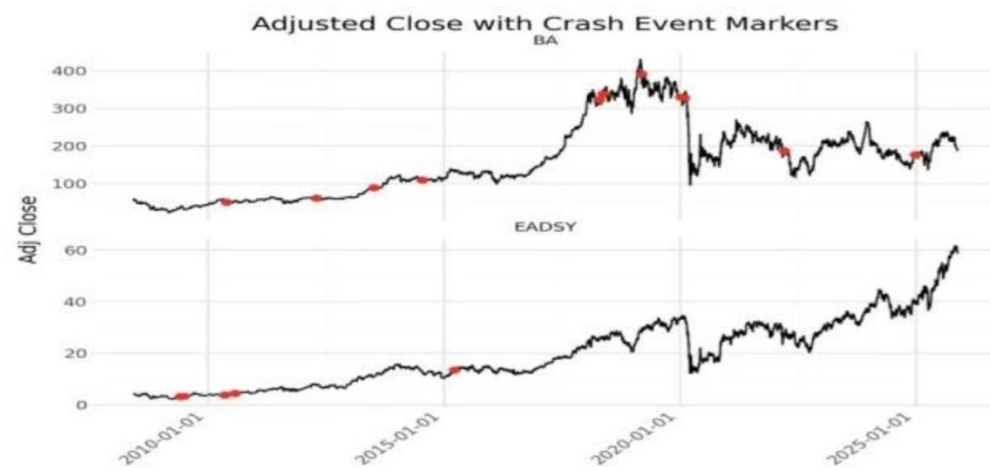
Why: Crash effects are not visible in raw prices because market movements, volatility, and unrelated news overwhelm the crash impact. This is why we need event-study methodology.

How: Plotted adjusted close prices over time and overlaid crash events as markers.



Price Paths with Crash Markers

For each stock, we plot Adjusted Close over time with crash days marked (red dots). We highlight large-severity crashes (>100 injuries) for readability.



Message: Price paths show local wobbles around crash dates, but not always huge breaks. This demonstrates the need for abnormal return analysis to isolate crash effects.

Event-Study Setup

Three Key Metrics

CAR

Cumulative Abnormal Return

Total impact of the crash on stock performance

TTR

Time to Recovery

How long the stock takes to return to pre-crash levels

MAE

Maximum Adverse Excursion

Worst drop after the crash within the event window

Together: CAR shows impact trend | TTR shows recovery speed | MAE shows worst moment



CAR: Cumulative Abnormal Return

Description

Daily total % gains/losses of the stock relative to the market index over a short time window

How It Works

1. Perform linear regression on market data -120 to -20 days of event
2. Use trend line to compute relative returns of the stock each day between -5 to +20 days of event
3. Accumulate those returns each day

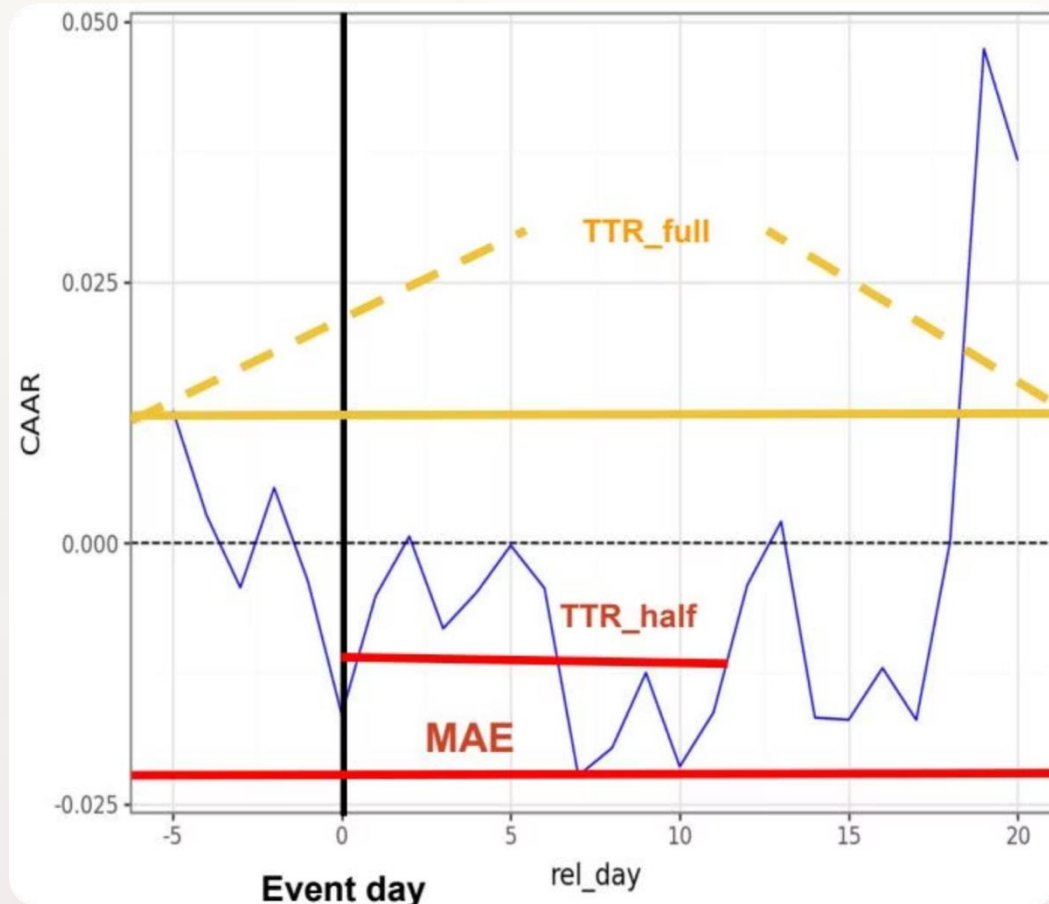
Why It Matters

- Remove market-wide noise from stock performance
- Isolate event impact on a specific stock
- Simplifies stock impact analysis

📄 **CAR > 0:** positive trend
(investors gain)

CAR < 0: negative trend
(investors lose)





MAE & TTR Explained



MAE (Worst CAR)

The most negative CAR during event window – represents the worst moment for investors



TTR_half

Time, in days, it took CAR to recover up to 50% of MAE



TTR_full

Time, in days, it took CAR to return to pre-event levels

Case Study: Air France AF447

Event Details

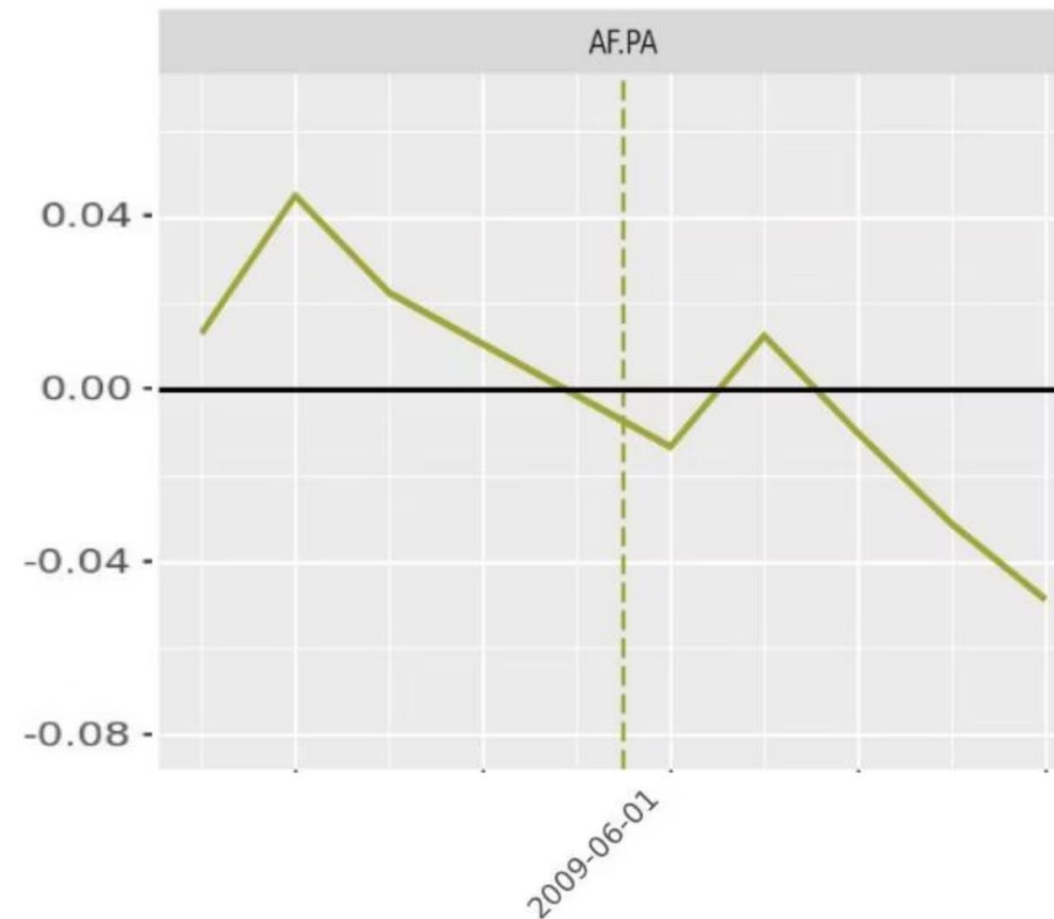
Flight: AF447 (Air France Airbus)

Date: 1 June 2009

Fatalities: 228

Market Impact

- Event-day CAR: -1% to -2%
- MAE: -5% to -7%
- TTR: 1-3 days (fast bounce)

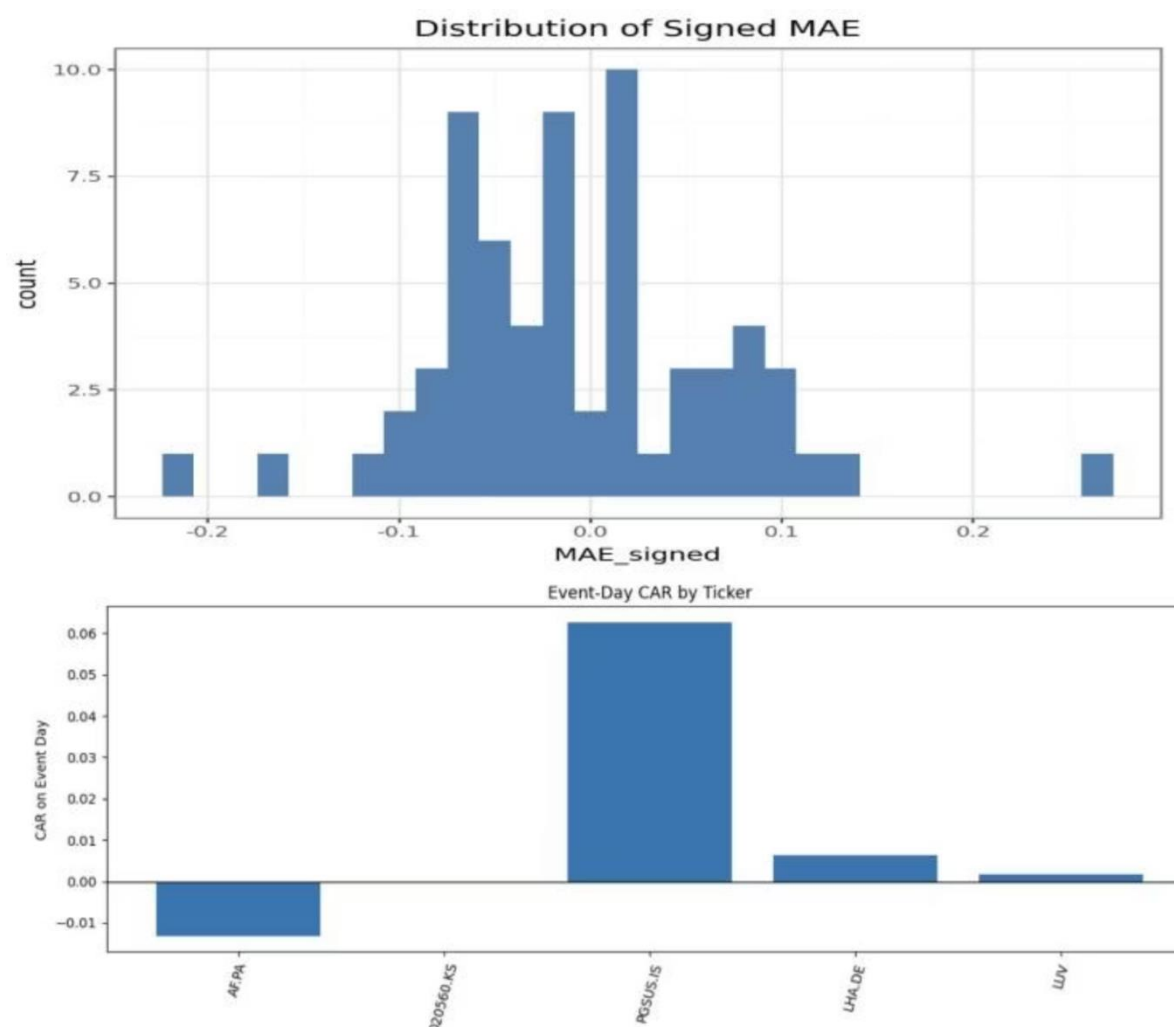


Takeaway: Even a global headline disaster produced a moderate, short-lived abnormal loss in the stock. This pattern is typical across our dataset.

How Big Are the Hits Overall?

Maximum Adverse Excursion (MAE) Analysis

Across 65 stock-event pairs with full windows, average worst abnormal move $\approx -3.3\%$ (mean CAR_min ≈ -0.033). Most events lie roughly between -10% and +5%; only a few beyond -15%.



Severe Case Zoom-In

For the 5 severe crashes with clean operator tickers (Air France, Asiana, Pegasus, Lufthansa, Southwest):

- Average MAE $\approx -3.5\%$
- Event-day CAR $\approx +1.4\%$ (some small rebounds)
- Average TTR ≈ 0.25 days (same/next trading day)

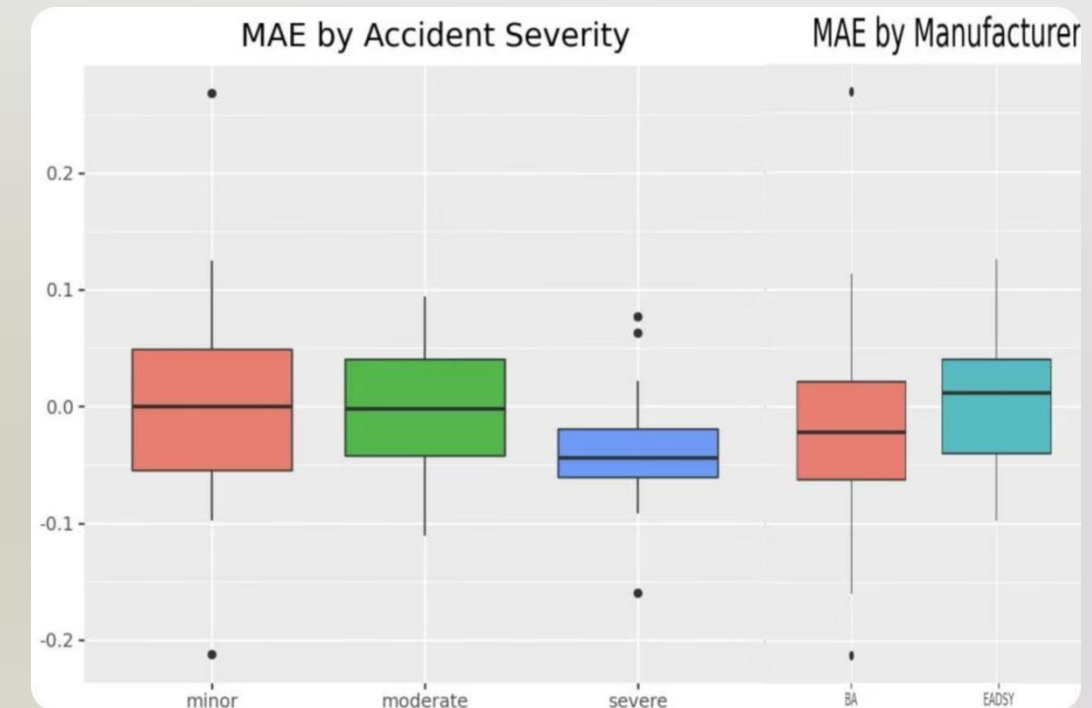
Does Severity or Manufacturer Matter?

By Severity

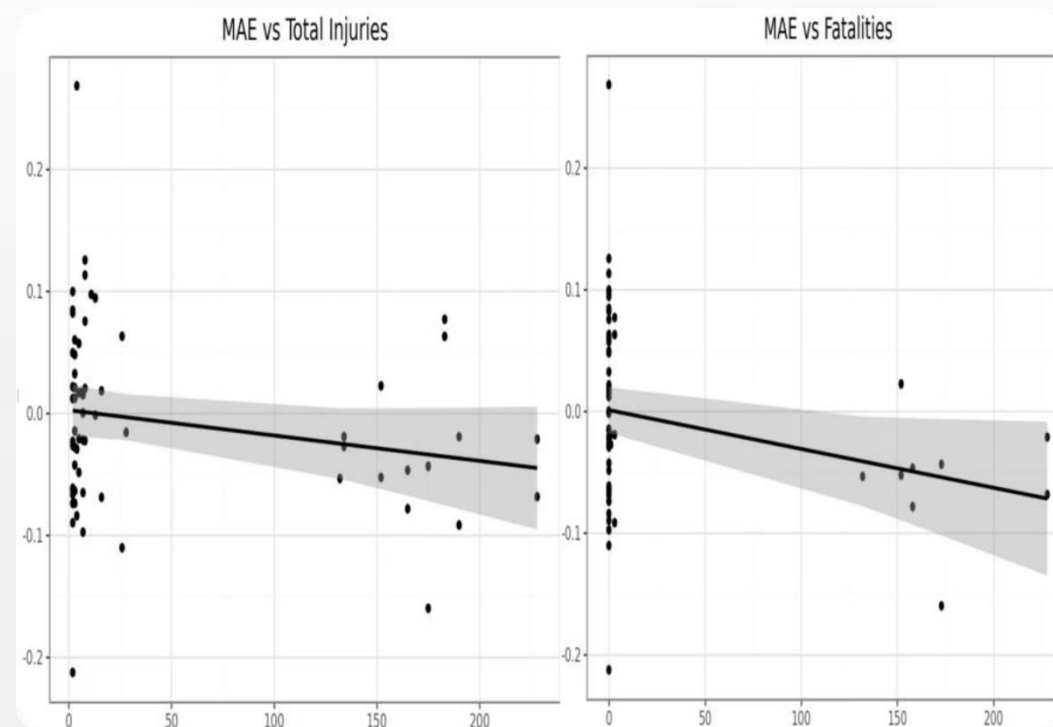
Minor, moderate, severe crashes have overlapping MAE distributions. Moderate events are only slightly more negative on median.

By Manufacturer

Boeing vs Airbus MAE boxplots are very similar → no strong evidence investors punish one maker more than the other, conditional on a crash.



Does the Body Count Matter?



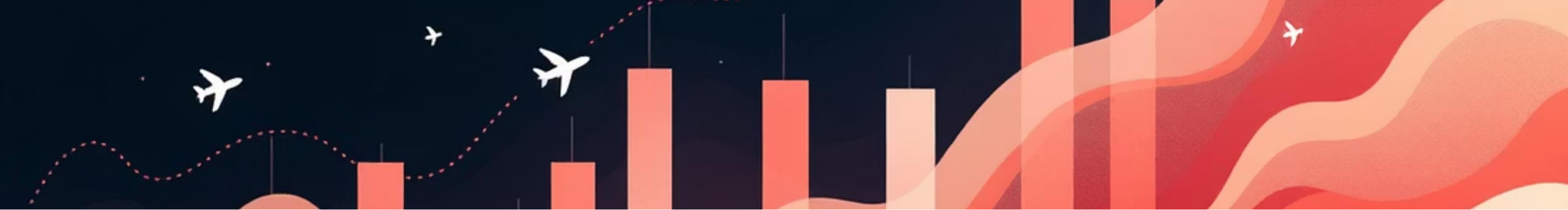
MAE vs Total Injuries

Slight downward trend, but weak and noisy with lots of spread

MAE vs Fatalities

Same story – no strong linear pattern emerges

Takeaway: Once an event is serious enough to be newsworthy, higher casualty counts don't proportionally increase the abnormal loss. The market reacts to the event itself, not the precise death toll.



Key Takeaways

Our analysis reveals distinct patterns in accident characteristics and their financial implications on public markets.

Accident Characteristics

- Most accidents since 2008 are minor; severe events are rare.
- Boeing and Airbus dominate accident counts, with Boeing involved in over twice as many as Airbus.
- US airlines show the highest counts due to market-listing bias and limited reporting from foreign carriers.

Market Abnormal Effects (MAE)

- Severe accidents show entirely negative MAE; overall distribution is strongly skewed negative.
- Boeing and Airbus have similar average MAE, though Boeing's is slightly more negative.
- MAE declines linearly as casualties increase in severe accidents.

Time to Recovery (TTR)

- Most accidents show TTR > 21 days, indicating no full recovery under model assumptions.
- Boeing shows faster recovery (lower TTR) than Airbus, despite more incidents.
- Higher severity and increased fatalities both correlate with longer TTR.

Cumulative Abnormal Returns (CAR)

- CAR drops sharply negative at the event date, then gradually increases but remains negative through the event window.
- Both Boeing and Airbus stocks exhibit similar CAR behavior following incidents.